1

$$P(T) = 1 - P(\bar{T})$$

$$= 1 - \left(1 - \frac{1}{6}\right) \cdot \left(1 - \frac{1}{3}\right) \cdot \left(1 - \frac{7}{10}\right)$$

$$= 1 - \frac{5}{6} \cdot \frac{2}{3} \cdot \frac{3}{10} = 1 - \frac{1}{6} = \frac{5}{6}$$

2

$$P(D) = P(B \cap C) = p_b \cdot p_c$$

$$P(S) = P(D \cup A) = p_a + p_b \cdot p_c - p_a \cdot p_b \cdot p_c$$

3

$N = 10$

$G = 6$

$S = 4$

$Z = 3$

a)

$$\frac{\binom{4}{0}\binom{6}{3}}{\binom{10}{3}}$$

$$\binom{4}{0} = 1$$

$$\frac{6 \cdot 5 \cdot 4}{1 \cdot 2 \cdot 3} = 5 \cdot 4$$

$$\frac{10 \cdot 9 \cdot 8}{1 \cdot 2 \cdot 3} = 5 \cdot 3 \cdot 8$$

$$= \frac{5 \cdot 4}{5 \cdot 3 \cdot 8_2} = \frac{1}{6}$$

b)

$$\frac{\binom{4}{3}\binom{6}{0}}{\binom{10}{3}}$$

$$\binom{6}{0} = 1$$

$$\frac{4 \cdot 3 \cdot 2}{1 \cdot 2 \cdot 3} = 4$$

$$\frac{4}{5 \cdot 3 \cdot 8_2} = \frac{1}{30}$$

4

$T \in [1, \infty)$

a

b

$$a \cdot \int_1^\infty \frac{1}{t^4}\, dt \overset{!}{=} 1$$

$$a \cdot \left(\frac{1}{-3t^3}\right)\Big|_1^\infty$$

$$a \cdot \left(0 - \frac{1}{-1^3}\right) =$$

$$a \cdot \frac{1}{3} \neq 1$$

$$\frac{a}{3} = 1$$

$$a = 3$$

$$f_{T(t)} = \begin{cases} 3\frac{1}{t^4} & , t \geq 1 \\ 0 & , t < 1 \end{cases}$$

c

$$F_{T(t)} = \int_{-\infty}^{t} f_{T(x)}\, dx$$

$$= \int_{-\infty}^{1} f_{T(x)}\, dx + \int_1^x f_{T(x)}\, dx$$

$$0 + \int_1^x 3\frac{1}{x^4}\, dx$$

$$= \left(-\frac{3}{3}\ \frac{1}{x^3}\right)\Big|_1^x$$

$$\frac{-1}{t^3} - \frac{1}{-1^3} = \underline{1 - \frac{1}{t^3}}$$

$$F_{T(t)} = \begin{cases} 1 - \frac{1}{t^3} & , t \geq 1 \\ 0 & , t < 1 \end{cases}$$

d

$$1 - \frac{1}{m^3} = \frac{1}{2}$$

$$1 - \frac{1}{2} = \frac{1}{m^3}$$

$$\frac{1}{2} = \frac{1}{m^3}$$

$$m^3 = 2$$

$$m = \sqrt[3]{2} = \underline{1{,}26}$$

e

$$E[T] = \int_{-\infty}^{\infty} x f_{T(x)}\, dx$$

$$= \int_1^\infty x \frac{3}{x^4}\, dx$$

$$= \int_1^\infty \frac{3}{x^3}\, dx$$

$$= \frac{3}{4}\left(-\frac{1}{x^4}\right)\Big|_1^\infty$$

$$\frac{3}{4}\left(0 - \frac{1}{-1^4}\right) = \frac{3}{4}$$

(5) $\lambda = \frac{1}{8a}$ (a) $F_{T(t)} = 1 - e^{-\lambda t}$

$t = 2a$

$= 1 - e^{-\frac{1}{8a} \cdot 2a}$

$= 1 - e^{-\frac{1}{4}} = 1 - 0{,}779 = 0{,}221$

(b) $E[T] = \frac{1}{\lambda} = \frac{1}{\frac{1}{8a}} = 8a$

(6)

Normalverteilung, Gleichverteilung, Exponentialverteilung
Gammaverteilung, Weibullverteilung

(7) Gleichverteilung, Binomialverteilung, Poission Verteilung
Hypergeometrische Verteilung, Geometrische Verteilung, Bernoulli Verteilung

(8)

$S^2 = \frac{1}{N-1} \sum_{i=0}^{N} (t_i - \bar{t})^2$

(9)

$S =$

(10)

(11)

2 x pro Monat relativ selten → Possion Verteilung

4 x pro 2 Monate

$P(X<3) = P(X \leq 2)$

$\lambda = 4$

$= e^{-\lambda} \sum_{k=0}^{2} \frac{\lambda^k}{k!} = e^{-4} \left(\frac{4^0}{1} + \frac{4^1}{1} + \frac{4^2}{2} \right)$

$e^{-4} \left(1 + 4 + \frac{4 \cdot 4^2}{2} \right)$

$= e^{-4} \cdot 13 = 0{,}238 \rightarrow 23{,}8\%$

(12)

$S = X + Y$

$\quad N(\mu_Y, \sigma_Y^2)$ $\qquad \mu_S = \mu_X + \mu_Y = 3 + 2 = 5$

$\quad N(\mu_X, \sigma_X^2)$ $\qquad \sigma_S^2 = \sigma_X^2 + \sigma_Y^2 = 4^2 + 3^2 = 5^2$

$\quad N(\mu_S + \sigma_S^2)$

$\rightarrow S = N(5, 5^2) =$

$P(\cancel{x} < 10) = \Phi\left(\frac{10-5}{5}\right) = \Phi(1) = 0{,}8419$

$\rightarrow \underline{84{,}19\%}$

(13)

$N = 100$

$100 \cdot 4s = 400s$

$8\,\text{min} = 8 \cdot 60s = 480s$

$\sigma = \sqrt{N \cdot \sigma^2} = \sqrt{100 \cdot 10^2} = 100$

$P(X \leq 480) = \Phi\left(\frac{480-400}{100}\right) = \Phi(0{,}8)$

$= 0{,}7881$

$\rightarrow \underline{78{,}81\%}$

(14)

$N = 9$

$\bar{A} = 50$

$\alpha = 0{,}01$

$S^2 = 900$

$\sigma = \sqrt{S} = \sqrt{900} = 30$

$Z_{-\frac{\alpha}{2}} \rightarrow 1 - \alpha = 0{,}99$

$\rightarrow$ Tabelle (∞)

$= 2{,}376$

$P\left(\bar{A} - Z_{-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{N}} \leq \mu \leq \bar{A} + Z_{-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{N}}\right)$

$P\left(50 - 2{,}376 \underbrace{\frac{6}{\sqrt{9}}}_{2} \leq \mu \leq 50 + 2{,}376 \frac{6}{\sqrt{9}}\right)$

$P(50 - 4{,}752 \leq \mu \leq 50 + 4{,}752)$

$P(\underline{45{,}248} \leq \mu \leq \underline{54{,}752})$